VCE GENERAL MATHEMATICS · PRACTICE EXAM

General Mathematics · Medium paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ **Date** _____

Exam 1 · multiple choice	Exam 2 · written response	State average on these questions
40 marks	**60 marks**	**58 / 100 (58%)**
reading 15 min · writing 1 h 30 min	reading 15 min · writing 1 h 30 min	a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright.
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Working can be typed in plain text, e.g. 0.004 × 5440.88 = 21.76.
3. Keep to the real time limits above, with your CAS, your bound reference and the VCAA formula sheet, and don't look at the guide until you've finished.
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5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question from the 2023–2025 General Mathematics exams, chosen by ranking all the questions by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions) within each area of study. This paper takes the middle band, with the real exam's balance of areas.

Sources

- 2023: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2024: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2025: Exam 1 (PDF) · report · Exam 2 (PDF) · report

Exam 1 · multiple choice (40 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then click your answer. 2023 questions have five options (A–E); 2024 and 2025 questions have four (A–D).

Data analysis

Question 1 · 1 mark

2023 VCAA Exam 1 · Question 12 · open in the official exam PDF (p. 9) ↗

Age of a tree from a $\log_{10}$(age) model and a given height

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 2 · 1 mark

2023 VCAA Exam 1 · Question 13 · open in the official exam PDF (p. 10) ↗

Seven-median smoothed winning time for 2006 from a time series plot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 3 · 1 mark

2023 VCAA Exam 1 · Question 15 · open in the official exam PDF (p. 11) ↗

Eight-mean smoothed value with centring for day 6 of library visitors

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 4 · 1 mark

2024 VCAA Exam 1 · Question 1 · open in the official exam PDF (p. 3) ↗

Percentage choosing blue, read from a percentage segmented bar chart

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 5 · 1 mark

2024 VCAA Exam 1 · Question 3 · open in the official exam PDF (p. 4) ↗

Frequency of the column containing the median of a log-scale histogram of 27 countries

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 6 · 1 mark

2024 VCAA Exam 1 · Question 4 · open in the official exam PDF (p. 4) ↗

Which value could be the upper outlier on the log-scale histogram

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 7 · 1 mark

2024 VCAA Exam 1 · Question 8 · open in the official exam PDF (p. 6) ↗

Least squares line for average female athletes against Games number, from a scatterplot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 8 · 1 mark

2024 VCAA Exam 1 · Question 10 · open in the official exam PDF (p. 7) ↗

Which Games has a predicted average number of male athletes of 25.6

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 9 · 1 mark

2024 VCAA Exam 1 · Question 13 · open in the official exam PDF (p. 9) ↗

Seven-median smoothed number of new staff for 2016

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 10 · 1 mark

2024 VCAA Exam 1 · Question 14 · open in the official exam PDF (p. 9) ↗

New staff in 2023, given the mean over 13 years

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 11 · 1 mark

2024 VCAA Exam 1 · Question 15 · open in the official exam PDF (p. 10) ↗

Six-mean smoothed value with centring for month 5 of cans sold

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 12 · 1 mark

2025 VCAA Exam 1 · Question 1 · open in the official exam PDF (p. 2) ↗

Median bag size from a histogram of avocado bag sizes

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 13 · 1 mark

2025 VCAA Exam 1 · Question 2 · open in the official exam PDF (p. 2) ↗

Total number of avocados sold, from the same histogram

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 14 · 1 mark

2025 VCAA Exam 1 · Question 9 · open in the official exam PDF (p. 6) ↗

Least squares line for games won against goals against, read from a scatterplot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 15 · 1 mark

2025 VCAA Exam 1 · Question 10 · open in the official exam PDF (p. 6) ↗

What r = −0.466 between goals against and games won lets you conclude

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 16 · 1 mark

2025 VCAA Exam 1 · Question 13 · open in the official exam PDF (p. 8) ↗

Five-median smoothed margin for week 8 from a time series plot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Recursion and financial modelling

Question 17 · 1 mark

2023 VCAA Exam 1 · Question 21 · open in the official exam PDF (p. 13) ↗

Reducing balance rate for the same computer

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 18 · 1 mark

2024 VCAA Exam 1 · Question 21 · open in the official exam PDF (p. 12) ↗

Total interest on a five-year loan with given monthly repayments

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 19 · 1 mark

2024 VCAA Exam 1 · Question 23 · open in the official exam PDF (p. 13) ↗

Time to repay the same loan

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 20 · 1 mark

2025 VCAA Exam 1 · Question 17 · open in the official exam PDF (p. 10) ↗

Calculation for the balance of a simple interest investment after three years

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 21 · 1 mark

2025 VCAA Exam 1 · Question 18 · open in the official exam PDF (p. 10) ↗

Type of sequence when R = 0.5 and d = 0 with a positive start

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 22 · 1 mark

2025 VCAA Exam 1 · Question 19 · open in the official exam PDF (p. 10) ↗

When flat rate first falls below reducing balance, from a table of values

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 23 · 1 mark

2025 VCAA Exam 1 · Question 20 · open in the official exam PDF (p. 11) ↗

Unit cost depreciation per hour from values two years apart

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 24 · 1 mark

2025 VCAA Exam 1 · Question 24 · open in the official exam PDF (p. 12) ↗

Total length of an annuity whose payment drops part-way through

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Matrices

Question 25 · 1 mark

2023 VCAA Exam 1 · Question 28 · open in the official exam PDF (p. 17) ↗

Which dominance matrix matches a given round-robin ranking

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 26 · 1 mark

2023 VCAA Exam 1 · Question 31 · open in the official exam PDF (p. 18) ↗

Meaning of the element in row 2, column 1 of a Leslie matrix

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 27 · 1 mark

2024 VCAA Exam 1 · Question 26 · open in the official exam PDF (p. 15) ↗

Expression for prices after a 15% discount

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 28 · 1 mark

2024 VCAA Exam 1 · Question 30 · open in the official exam PDF (p. 18) ↗

Leslie matrix from a table of birth and survival rates

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 29 · 1 mark

2025 VCAA Exam 1 · Question 27 · open in the official exam PDF (p. 13) ↗

Which pair of values of m and n makes a 2 × 2 matrix singular

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 30 · 1 mark

2025 VCAA Exam 1 · Question 30 · open in the official exam PDF (p. 16) ↗

Number of negative elements when $f_{ij} = i^2 - j$ in a 4 × 4 matrix

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 31 · 1 mark

2025 VCAA Exam 1 · Question 31 · open in the official exam PDF (p. 16) ↗

Which matrix computation is defined, given four matrices of different orders

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 32 · 1 mark

2025 VCAA Exam 1 · Question 32 · open in the official exam PDF (p. 17) ↗

Complete a partial dominance matrix from match information and find $T = D + D^2$

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Networks and decision mathematics

Question 33 · 1 mark

2023 VCAA Exam 1 · Question 33 · open in the official exam PDF (p. 20) ↗

How many of five statements about a small graph are true

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 34 · 1 mark

2023 VCAA Exam 1 · Question 35 · open in the official exam PDF (p. 21) ↗

Weight of the minimum spanning tree of a small weighted graph

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 35 · 1 mark

2023 VCAA Exam 1 · Question 37 · open in the official exam PDF (p. 22) ↗

Number of faces of a planar graph given its adjacency matrix

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 36 · 1 mark

2023 VCAA Exam 1 · Question 38 · open in the official exam PDF (p. 23) ↗

Which activity network (with a dummy) matches a precedence table

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 37 · 1 mark

2023 VCAA Exam 1 · Question 39 · open in the official exam PDF (p. 24) ↗

Capacity of the minimum cut of a maze flow network

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 38 · 1 mark

2025 VCAA Exam 1 · Question 35 · open in the official exam PDF (p. 19) ↗

Length of a minimum spanning tree that contains an edge of unknown weight w

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 39 · 1 mark

2025 VCAA Exam 1 · Question 37 · open in the official exam PDF (p. 20) ↗

Maximum flow from source to sink in the same network

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 40 · 1 mark

2025 VCAA Exam 1 · Question 39 · open in the official exam PDF (p. 22) ↗

Table after row and column reduction (Hungarian algorithm) of a contractor cost table

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Exam 2 · written response (60 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Data analysis

Question 1 (9 marks)

2023 VCAA Exam 2 · Question 1 · open in the official exam PDF (p. 2) ↗

Oyster data (weight, volume, image size, size) for 15 oysters: summary statistics, a least squares line and the normal model

a. Number of categorical variables in the table (1 mark)

b.i. Mean weight of all 15 oysters (1 mark)

b.ii. Median weight of the large oysters only (1 mark)

c.i. Response variable in the given volume–weight equation (1 mark)

c.ii. Predicted change in volume for a 10 g increase in weight (1 mark)

d. Least squares line for volume against image size, coefficients to four significant figures (2 marks)

e. Mean and standard deviation of megapixels from two percentages, using the 68–95–99.7% rule (2 marks)

Question 2 (8 marks)

2024 VCAA Exam 2 · Question 1 · open in the official exam PDF (p. 2) ↗

Men's Olympic high jump gold medal heights (Mgold), 1928–2020: summary statistics, z-score, boxplot and a least squares line against year

a.i. Maximum Mgold (1 mark)

a.ii. Percentage of Mgold values greater than 2.25 m (1 mark)

b. Standardised score for the 2000 height of 2.35 m (mean 2.23, sd 0.15) (1 mark)

c. Construct a boxplot of Mgold on a grid (2 marks)

d. Least squares line for Mgold against year, coefficients to three significant figures (2 marks)

e. Interpret $r^2 = 0.857$ (1 mark)

Question 3 (3 marks)

2025 VCAA Exam 2 · Question 5 · open in the official exam PDF (p. 8) ↗

Apartment sale price against days on the market for 10 apartments

a. Least squares line, both values to four significant figures (2 marks)

b. Meaning of the coefficient of determination as a whole percentage, from $r = -0.866$ (1 mark)

Question 4 (4 marks)

2025 VCAA Exam 2 · Question 6 · open in the official exam PDF (p. 9) ↗

Monthly home sales over four years: qualitative features and a seasonal index for September

a. Two qualitative features of the time series (excluding outliers) (2 marks)

b. Seasonal index for September to three decimal places, from yearly totals (2 marks)

Recursion and financial modelling

Question 5 (4 marks)

2023 VCAA Exam 2 · Question 6 · open in the official exam PDF (p. 11) ↗

A $600 000 annuity paying $3973 per month: interest rate, amortisation table and recurrence relation

a. Annual interest rate from 0.42% per month (1 mark)

b. Complete the next line of the amortisation table (1 mark)

c. Recurrence relation for the annuity balance (1 mark)

d. Name for an annuity whose balance stays constant (1 mark)

Question 6 (4 marks)

2024 VCAA Exam 2 · Question 7 · open in the official exam PDF (p. 12) ↗

A \$300 000 annuity with $E_{n+1} = 1.003E_n - 2159.41$

a. Balance after two months, showing recursive calculations (1 mark)

b. Number of years of regular payments (1 mark)

c. Annual compound interest rate (1 mark)

d. Monthly payment if it were a perpetuity (1 mark)

Question 7 (4 marks)

2025 VCAA Exam 2 · Question 7 · open in the official exam PDF (p. 10) ↗

A reducing balance loan with monthly repayments of $15 730.88: amortisation table

a. Amount borrowed (1 mark)

b. Why the interest for payment 2 is lower than for payment 1 (1 mark)

c. Complete the payment 3 row from the table values (1 mark)

d. Number of full payments before a final payment of $15 730.71 (1 mark)

Matrices

Question 8 (3 marks)

2023 VCAA Exam 2 · Question 8 · open in the official exam PDF (p. 14) ↗

Circus ticket prices in a column matrix N (family, adult, child)

a. Which element gives the child ticket price? (1 mark)

b. Complete a matrix equation showing a family ticket saves $10 (1 mark)

c. Matrix K that gives revenue for each ticket type from N (1 mark)

Question 9 (3 marks)

2023 VCAA Exam 2 · Question 9 · open in the official exam PDF (p. 15) ↗

Circus ticket sales at five locations (row matrix R) and a permutation matrix for the touring order

a. Scalar that gives average sales per performance (20 performances) (1 mark)

b. Value of t for a 25% increase in revenue (1 mark)

c. Order of locations visited, starting from I, from the permutation matrix (1 mark)

Question 10 (3 marks)

2023 VCAA Exam 2 · Question 10 · open in the official exam PDF (p. 17) ↗

A communication matrix G between circus groups (directors, managers, performers, sales staff, customers)

a. Shortest communication sequence for a customer complaint to reach a director (1 mark)

b.i. Complete the two-step matrix $H = G^2$ (1 mark)

b.ii. What g_{21} and h_{21} say about communication (1 mark)

Question 11 (3 marks)

2024 VCAA Exam 2 · Question 11 · open in the official exam PDF (p. 17) ↗

A Leslie matrix model of a native animal's female population in four age groups

a.i. Complete the life-cycle (transition) diagram (1 mark)

a.ii. Population of each age group after one year (1 mark)

b. Years until the total female population is first half its initial size (1 mark)

Networks and decision mathematics

Question 12 (3 marks)

2023 VCAA Exam 2 · Question 13 · open in the official exam PDF (p. 22) ↗

Roads between nine landmarks: shortest path, a Hamiltonian cycle and repeating roads to travel every road

a. Minimum distance from G to M (1 mark)

b. A route visiting every landmark and returning to G with minimum distance (1 mark)

c. The two roads travelled twice to cover every road, starting at G and ending elsewhere (1 mark)

Question 13 (3 marks)

2024 VCAA Exam 2 · Question 14 · open in the official exam PDF (p. 24) ↗

Deliveries from a manufacturer to a supermarket through warehouses: cut capacity, maximum flow and the best edge to increase

a. Capacity of the marked cut (1 mark)

b. Maximum deliveries per day (1 mark)

c. Pair of locations whose link should be increased (1 mark)

Question 14 (2 marks)

2025 VCAA Exam 2 · Question 16 · open in the official exam PDF (p. 24) ↗

An adjacency matrix for passages between five gym areas, with three unknown values

Values of x, y and z (2 marks)

Question 15 (4 marks)

2025 VCAA Exam 2 · Question 18 · open in the official exam PDF (p. 26) ↗

A 12-activity home gym project: two critical paths, latest start time, float and crashing

a. Activities common to both critical paths (1 mark)

b. Latest start time for E (1 mark)

c. Activity with the longest float (1 mark)

d. Minimum extra cost to finish three days sooner (1 mark)

Marking guide

Mark yourself only after you've finished both exams. Tick the boxes you earned, enter your marks and compare with how Victoria went on each question.

Exam 1 · multiple choice

✓	Q	Ans	Why	How Victoria went
☐	1	**D**	$\log_{10}$(age) = (8.52 + 3.8) ÷ 12.6 = 0.978, so age = 10^0.978 ≈ 9.5 years.	76% correct A 10% · B 6% · C 4% · D 76% · E 4%
☐	2	**C**	Take the seven years centred on 2006 (2003 to 2009) and find the middle value of their winning times: about 116.8 s.	68% correct A 4% · B 12% · C 68% · D 13% · E 3%
☐	3	**C**	Mean of days 2–9 is 323 and of days 3–10 is 327; centring gives (323 + 327) ÷ 2 = 325.	66% correct A 6% · B 5% · C 66% · D 9% · E 15%
☐	4	**B**	Blue is the segment from 56% to 74%: 74 − 56 = 18%.	78% correct A 18% · B 78% · C 3% · D 1%
☐	5	**D**	The median is the 14th value; counting from the left it falls in the column with frequency 9.	71% correct A 18% · B 5% · C 7% · D 71%
☐	6	**C**	The outlier lies between $\log_{10}$ = 3.2 and 3.4, i.e. between about 1585 and 2512. Only 2030 ($\log_{10}$ ≈ 3.31) fits.	68% correct A 11% · B 14% · C 68% · D 6%
☐	7	**B**	Read two points on the line and find slope ≈ 0.91 and intercept ≈ −3.39; females is the response variable.	73% correct A 17% · B 73% · C 5% · D 4%
☐	8	**C**	25.6 = 67.5 − 1.27 × number gives number ≈ 33: the 33rd Games.	73% correct A 6% · B 11% · C 73% · D 10%
☐	9	**A**	The seven values centred on 2016 (2013–2019), in order: 6, 6, 7, 10, 11, 12, 13. The median is 10.	72% correct A 72% · B 13% · C 7% · D 8%
☐	10	**B**	13 years with mean 11 total 143. The previous 12 years total 127, so 2023 had 16.	68% correct A 15% · B 68% · C 12% · D 5%
☐	11	**C**	The six-mean of months 2–7 is 318 and of months 3–8 is 324; centring gives 321.	70% correct A 7% · B 10% · C 70% · D 12%
☐	12	**D**	39 customers, so the median is the 20th value; counting up the columns it falls in bag size 4.	75% correct A 4% · B 5% · C 17% · D 75%
☐	13	**D**	Multiply each bag size by its frequency and add: 4 + 22 + 9 + 36 + 25 + 42 = 138.	76% correct A 20% · B 2% · C 1% · D 76%
☐	14	**A**	Use two points on the line, e.g. (27, 12) and (44, 9): slope ≈ −0.178 and intercept ≈ 16.8.	66% correct A 66% · B 6% · C 23% · D 4%
☐	15	**D**	Negative association: more goals against goes with fewer wins. Correlation doesn't show causation.	63% correct A 22% · B 7% · C 8% · D 63%
☐	16	**C**	Weeks 6 to 10 give 50, 38, 32, 35 and 41; in order the middle value is 38.	76% correct A 10% · B 8% · C 76% · D 5%
☐	17	**E**	3000 × R^4 = 600 gives R = 0.2^0.25 ≈ 0.669, a rate of about 33%.	48% correct A 7% · B 14% · C 20% · D 11% · E 48%
☐	18	**D**	Total repaid = 2228.40 × 60 = 133 704; interest = 133 704 − 121 000 = $12 704.	56% correct A 12% · B 15% · C 16% · D 56%

☐	19	**A**	I% = 0.4 × 12 = 4.8, PV = 240 000, Pmt = −2741.05, FV = 0: N = 108 months, which is 9 years.	54% correct A 54% · B 13% · C 18% · D 14%
☐	20	**D**	Simple interest adds 0.04 × 4000 each year: 4000 + 3(0.04 × 4000).	53% correct A 32% · B 6% · C 9% · D 53%
☐	21	**D**	Multiplying by 0.5 each time gives a geometric, decreasing sequence.	52% correct A 14% · B 15% · C 19% · D 52%
☐	22	**B**	Flat rate falls 4000 a year; reducing balance multiplies by 0.92. After 5 years: 40 000 vs 39 545; after 6: 36 000 vs 36 381. Year 6.	49% correct A 17% · B 49% · C 25% · D 8%
☐	23	**A**	(12 000 − 7680) ÷ (2 × 960) = $2.25 per hour.	55% correct A 55% · B 19% · C 7% · D 18%
☐	24	**A**	N for 6000 payments until the balance is 521 118.96 is 84; N for 4767.66 payments to 0 is 144. 228 months = 19 years.	52% correct A 52% · B 16% · C 17% · D 14%
☐	25	**D**	Row sums must give U 3 wins, V 2, S 1 and T 0, with one result per pair. Only D does this.	69% correct A 3% · B 5% · C 5% · D 69% · E 19%
☐	26	**C**	Elements just below the leading diagonal are survival rates: 20% survive from the first year into the second.	74% correct A 5% · B 8% · C 74% · D 10% · E 3%
☐	27	**B**	Each price is multiplied by 85%: 0.85C.	76% correct A 20% · B 76% · C 2% · D 2%
☐	28	**B**	Birth rates (0, 1.8, 1.2) go in row 1 and survival rates 0.7, 0.6 below the diagonal: matrix B.	78% correct A 5% · B 78% · C 7% · D 10%
☐	29	**C**	det = mn + 36 must not be 0, so mn ≠ −36: 3 and −12 is not allowed.	76% correct A 4% · B 6% · C 76% · D 13%
☐	30	**B**	Only row 1 has negatives (j = 2, 3, 4). Row 2 gives 4 − j ≥ 0. Three elements.	73% correct A 10% · B 73% · C 12% · D 5%
☐	31	**B**	A is 2 × 2, B 3 × 1, C 3 × 2, D 1 × 3. C^T(2 × 3) B(3 × 1) is 2 × 1, and times D(1 × 3) gives 2 × 3. The others fail.	70% correct A 5% · B 70% · C 14% · D 10%
☐	32	**B**	Kyle's two wins are Maggie and Neil, so Lian beat Kyle. Lian's only win is Kyle, so Neil beat Lian. Complete D, compute D^2 and add: matrix B.	63% correct A 10% · B 63% · C 6% · D 21%
☐	33	**D**	Check each property against the drawing: four of the five statements hold.	56% correct A 3% · B 11% · C 30% · D 56% · E 1%
☐	34	**C**	Use Prim's algorithm: the cheapest connecting edges total 40.	74% correct A 3% · B 3% · C 74% · D 12% · E 7%
☐	35	**B**	The matrix total is 20, so there are 10 edges. Euler: 5 + f = 10 + 2, so f = 7.	66% correct A 7% · B 66% · C 14% · D 5% · E 7%
☐	36	**D**	F and I both follow D and E, so D and E end at one vertex where F and I start. G needs C and F but H needs only F, so a dummy joins the end of F to the start of G. Only option D shows exactly this.	60% correct A 13% · B 10% · C 10% · D 60% · E 6%
☐	37	**B**	Check cuts from entrance to exit, counting only edges from the entrance side to the exit side: the smallest is 23.	62% correct A 6% · B 62% · C 12% · D 6% · E 14%
☐	38	**B**	Prim's algorithm on the other edges, plus w: 5 + 6 + 4 + 2 + 4 + 5 + 3 + 6 + 5 + 4 + w = 44 + w.	65% correct A 8% · B 65% · C 15% · D 12%

☐	39	**B**	The minimum cut is through the three pipes into the sink: 7 + 4 + 6 = 17.	73% correct A 13% · B 73% · C 10% · D 3%
☐	40	**B**	Row minimums 9600, 10 000, 12 400, 9000, then column minimums 4500, 0, 13 150, 8700: table B.	71% correct A 8% · B 71% · C 11% · D 10%

Multiple-choice score: ☐ **/ 40** **(state average 26.6)**

Exam 2 · written response

Question 1 (9 marks) · state average 4.6 / 9 (51%)

2023 VCAA Exam 2 · Question 1

a. Number of categorical variables in the table

☐ 2 (ID and size)

Answer / notes: 2. ID is a label and size is small/medium/large. Half the state missed it.

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: ☐ **/ 1**

b.i. Mean weight of all 15 oysters

☐ 11.42 g (exact; don't round)

Answer / notes: 11.42 g. Rounding to 11.4 lost the mark: no rounding instruction was given.

State average 0.9 / 1 (90%); 86% scored full marks. Mark distribution: 0: 14% · 1: 86%

Your mark: ☐ **/ 1**

b.ii. Median weight of the large oysters only

☐ 14.1 g

Answer / notes: 14.1 g. Many gave the median of the whole data set instead.

State average 0.4 / 1 (40%); 39% scored full marks. Mark distribution: 0: 61% · 1: 39%

Your mark: ☐ **/ 1**

c.i. Response variable in the given volume–weight equation

☐ volume

Answer / notes: Volume (the variable on the left of the equation).

State average 0.9 / 1 (90%); 88% scored full marks. Mark distribution: 0: 12% · 1: 88%

Your mark: ☐ **/ 1**

c.ii. Predicted change in volume for a 10 g increase in weight

☐ 9.53 cm^3

Answer / notes: 10 × 0.953 = 9.53 cm^3. A common error was 0.953 (one gram, not ten).

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: ☐ **/ 1**

d. Least squares line for volume against image size, coefficients to four significant figures

☐ Correct coefficients: 0.002857 and 2.571

☐ Correct placement: volume = 0.002857 + 2.571 × image size

Answer / notes: volume = 0.002857 + 2.571 × image size. Examiners saw coefficients wrongly rounded and variables swapped.

State average 1.0 / 2 (50%); 37% scored full marks. Mark distribution: 0: 38% · 1: 26% · 2: 37%

Your mark: ☐ **/ 2**

e. Mean and standard deviation of megapixels from two percentages, using the 68–95–99.7% rule

- [ ] Set up $\mu + 2\sigma = 4.6$ and $\mu - \sigma = 4.3$
- [ ] Mean 4.4, standard deviation 0.1

Answer / notes: 97.5% below 4.6 means $4.6 = \mu + 2\sigma$; 84% above 4.3 means $4.3 = \mu - \sigma$. So $\sigma = 0.1$ and $\mu = 4.4$.

State average 0.6 / 2 (30%); 29% scored full marks. Mark distribution: 0: 67% · 1: 3% · 2: 29%

Your mark: ☐ **/ 2**

Question 2 (8 marks) · state average 5.9 / 8 (74%)

2024 VCAA Exam 2 · Question 1

a.i. Maximum Mgold

- [ ] 2.39 m

Answer / notes: 2.39 m.

State average 1.0 / 1 (100%); 96% scored full marks. Mark distribution: 0: 4% · 1: 96%

Your mark: ☐ **/ 1**

a.ii. Percentage of Mgold values greater than 2.25 m

- [ ] 50%

Answer / notes: 50%. The most common wrong answer, 54.54%, included 2.25 itself.

State average 0.8 / 1 (80%); 77% scored full marks. Mark distribution: 0: 23% · 1: 77%

Your mark: ☐ **/ 1**

b. Standardised score for the 2000 height of 2.35 m (mean 2.23, sd 0.15)

- [ ] $z = 0.8$

Answer / notes: $(2.35 - 2.23) \div 0.15 = 0.8$.

State average 0.9 / 1 (90%); 84% scored full marks. Mark distribution: 0: 16% · 1: 84%

Your mark: ☐ **/ 1**

c. Construct a boxplot of Mgold on a grid

- [ ] Box from 2.12 to 2.36 with median 2.29
- [ ] Whiskers to 1.94 and 2.39

Answer / notes: Five-number summary 1.94, 2.12, 2.29, 2.36, 2.39. A complete boxplot with three correct values earned one mark.

State average 1.7 / 2 (85%); 76% scored full marks. Mark distribution: 0: 6% · 1: 17% · 2: 76%

Your mark: ☐ **/ 2**

d. Least squares line for Mgold against year, coefficients to three significant figures

- [ ] Intercept −7.97
- [ ] Slope 0.00516

Answer / notes: Mgold = −7.97 + 0.00516 × year. Many couldn't read CAS exponent notation (5.16E−3).

State average 1.0 / 2 (50%); 39% scored full marks. Mark distribution: 0: 35% · 1: 26% · 2: 39%

Your mark: ☐ **/ 2**

e. Interpret $r^2 = 0.857$

- [ ] 85.7% of the variation in Mgold can be explained by the variation in year

Answer / notes: Mention both variables, and don't round (86% lost the mark).

State average 0.5 / 1 (50%); 52% scored full marks. Mark distribution: 0: 48% · 1: 52%

Your mark: ☐ **/ 1**

Question 3 (3 marks) · state average 1.5 / 3 (50%)

2025 VCAA Exam 2 · Question 5

a. Least squares line, both values to four significant figures

- [ ] 1 050 000
- [ ] −8050

Answer / notes: sale price = 1 050 000 − 8050 × days. CAS shows 1.05E6: read exponent notation carefully.

State average 1.0 / 2 (50%); 28% scored full marks. Mark distribution: 0: 30% · 1: 43% · 2: 28%

Your mark: ____ **/ 2**

b. Meaning of the coefficient of determination as a whole percentage, from r = −0.866

- [ ] 75% of the variation in sale price can be explained by the variation in days

Answer / notes: $r^2 = 0.75$, so 75%.

State average 0.5 / 1 (50%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

Your mark: ____ **/ 1**

Question 4 (4 marks) · state average 1.5 / 4 (38%)

2025 VCAA Exam 2 · Question 6

a. Two qualitative features of the time series (excluding outliers)

- [ ] seasonality
- [ ] irregular fluctuations

Answer / notes: Seasonality and irregular fluctuations; there's no trend.

State average 1.1 / 2 (55%); 32% scored full marks. Mark distribution: 0: 23% · 1: 45% · 2: 32%

Your mark: ____ **/ 2**

b. Seasonal index for September to three decimal places, from yearly totals

- [ ] Each year: September value ÷ (yearly total ÷ 12)
- [ ] Average of the four proportions = 0.588

Answer / notes: 15 ÷ 30.083, 20 ÷ 29.5, 20 ÷ 29.833 and 15 ÷ 29.75, then average: 0.588. Only 12% got both marks.

State average 0.4 / 2 (20%); 12% scored full marks. Mark distribution: 0: 74% · 1: 14% · 2: 12%

Your mark: ____ **/ 2**

Question 5 (4 marks) · state average 2.1 / 4 (53%)

2023 VCAA Exam 2 · Question 6

a. Annual interest rate from 0.42% per month

- [ ] 5.04%

Answer / notes: 0.42 × 12 = 5.04%.

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

Your mark: ____ **/ 1**

b. Complete the next line of the amortisation table

- [ ] 3973.00, 2507.77, 1465.23, 595 622.67

Answer / notes: Interest = 0.0042 × 597 087.90 = 2507.77; principal reduction = 3973 − 2507.77 = 1465.23; balance = 595 622.67. Early rounding was common.

State average 0.4 / 1 (40%); 40% scored full marks. Mark distribution: 0: 60% · 1: 40%

Your mark: ____ **/ 1**

c. Recurrence relation for the annuity balance

- [ ] $V_0 = 600\,000$, $V_{n+1} = 1.0042V_n - 3973$

Answer / notes: $V_0 = 600\,000$, $V_{n+1} = 1.0042V_n - 3973$.

State average 0.4 / 1 (40%); 41% scored full marks. Mark distribution: 0: 59% · 1: 41%

Your mark: ____ **/ 1**

d. Name for an annuity whose balance stays constant

- [ ] perpetuity

Answer / notes: A perpetuity.

State average 0.6 / 1 (60%); 59% scored full marks. Mark distribution: 0: 41% · 1: 59%

Your mark: ____ **/ 1**

Question 6 (4 marks) · state average 2 / 4 (50%)

2024 VCAA Exam 2 · Question 7

a. Balance after two months, showing recursive calculations

- [ ] $E_1 = 298\,740.59$, $E_2 = 297\,477.40$

Answer / notes: $E_2 = \$297\,477.40$ (297 477.4 isn't to the nearest cent).

State average 0.5 / 1 (50%); 49% scored full marks. Mark distribution: 0: 51% · 1: 49%

Your mark: ____ **/ 1**

b. Number of years of regular payments

- [ ] 15 years

Answer / notes: Finance Solver with I% = 3.6, PV = −300 000, Pmt = 2159.41, FV = 0 gives N = 180 months.

State average 0.4 / 1 (40%); 42% scored full marks. Mark distribution: 0: 58% · 1: 42%

Your mark: ____ **/ 1**

c. Annual compound interest rate

- [ ] 3.6%

Answer / notes: 0.3% × 12 = 3.6%.

State average 0.6 / 1 (60%); 56% scored full marks. Mark distribution: 0: 44% · 1: 56%

Your mark: ____ **/ 1**

d. Monthly payment if it were a perpetuity

- [ ] $900

Answer / notes: 0.003 × 300 000 = $900.

State average 0.5 / 1 (50%); 47% scored full marks. Mark distribution: 0: 53% · 1: 47%

Your mark: ____ **/ 1**

Question 7 (4 marks) · state average 2.5 / 4 (63%)

2025 VCAA Exam 2 · Question 7

a. Amount borrowed

- [ ] $850 000

Answer / notes: $850 000 (balance at payment 0).

State average 1.0 / 1 (100%); 93% scored full marks. Mark distribution: 0: 7% · 1: 93%

Your mark: ____ **/ 1**

b. Why the interest for payment 2 is lower than for payment 1

- [ ] It is calculated on a lower balance

Answer / notes: Interest is charged on the balance, which has fallen.

State average 0.7 / 1 (70%); 65% scored full marks. Mark distribution: 0: 35% · 1: 65%

Your mark: ____ **/ 1**

c. Complete the payment 3 row from the table values

- [ ] 2885.55, 12 845.33, 811 598.26

Answer / notes: Interest = 0.0035 × 824 443.59 = 2885.55; principal reduction 12 845.33; balance 811 598.26. Using Finance Solver gave a slightly different balance, which wasn't accepted.

State average 0.4 / 1 (40%); 40% scored full marks. Mark distribution: 0: 60% · 1: 40%

Your mark: ☐ **/ 1**

d. Number of full payments before a final payment of $15 730.71

- [ ] 59

Answer / notes: N = 60 in Finance Solver, so 59 full payments then the final one.

State average 0.4 / 1 (40%); 41% scored full marks. Mark distribution: 0: 59% · 1: 41%

Your mark: ☐ **/ 1**

Question 8 (3 marks) · state average 1.2 / 3 (40%)

2023 VCAA Exam 2 · Question 8

a. Which element gives the child ticket price?

- [ ] n_{31}

Answer / notes: n_{31} (row 3, column 1), with a lower-case n. A capital N or reversed subscripts lost the mark.

State average 0.5 / 1 (50%); 48% scored full marks. Mark distribution: 0: 52% · 1: 48%

Your mark: ☐ **/ 1**

b. Complete a matrix equation showing a family ticket saves $10

- [ ] [0 2 2] × N − [1 0 0] × N = 10

Answer / notes: Two adults and two children cost 2 × 15 + 2 × 8 = 46, versus 36 for a family ticket.

State average 0.4 / 1 (40%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

Your mark: ☐ **/ 1**

c. Matrix K that gives revenue for each ticket type from N

- [ ] K = diagonal matrix with 204, 162, 176 on the leading diagonal

Answer / notes: K must be 3 × 3 with the ticket numbers on the diagonal and 0s elsewhere. A row or column matrix was the usual error.

State average 0.3 / 1 (30%); 34% scored full marks. Mark distribution: 0: 66% · 1: 34%

Your mark: ☐ **/ 1**

Question 9 (3 marks) · state average 1.5 / 3 (50%)

2023 VCAA Exam 2 · Question 9

a. Scalar that gives average sales per performance (20 performances)

- [ ] 1/20 (or 0.05) × R

Answer / notes: 0.05 × R. Many struggled with the scalar.

State average 0.3 / 1 (30%); 25% scored full marks. Mark distribution: 0: 75% · 1: 25%

Your mark: ☐ **/ 1**

b. Value of t for a 25% increase in revenue

- [ ] 1.25

Answer / notes: t = 1.25. A common error was 0.25.

State average 0.5 / 1 (50%); 47% scored full marks. Mark distribution: 0: 53% · 1: 47%

Your mark: ☐ **/ 1**

c. Order of locations visited, starting from I, from the permutation matrix

☐ I – H – E – G – F

Answer / notes: Follow each column (this month) to its 1 (next month): I → H → E → G → F.

State average 0.7 / 1 (70%); 68% scored full marks. Mark distribution: 0: 32% · 1: 68%

Your mark: ☐ **/ 1**

Question 10 (3 marks) · state average 1.7 / 3 (57%)

2023 VCAA Exam 2 · Question 10

a. Shortest communication sequence for a customer complaint to reach a director

☐ C – S – M – D

Answer / notes: Customers reach sales staff, sales staff reach managers, managers reach directors.

State average 0.7 / 1 (70%); 74% scored full marks. Mark distribution: 0: 26% · 1: 74%

Your mark: ☐ **/ 1**

b.i. Complete the two-step matrix $H = G^2$

☐ All 16 missing elements correct ($H = G^2$)

Answer / notes: $H = G^2$ = rows (1 2 1 2 2), (0 3 1 2 2), (1 0 1 1 1), (1 0 1 2 1), (0 1 0 0 1).

State average 0.7 / 1 (70%); 66% scored full marks. Mark distribution: 0: 34% · 1: 66%

Your mark: ☐ **/ 1**

b.ii. What g_{21} and h_{21} say about communication

☐ Managers can communicate directly with directors, but not through exactly one other person

Answer / notes: $g_{21} = 1$: managers talk directly to directors. $h_{21} = 0$: there's no two-step link from managers to directors.

State average 0.3 / 1 (30%); 25% scored full marks. Mark distribution: 0: 75% · 1: 25%

Your mark: ☐ **/ 1**

Question 11 (3 marks) · state average 1.3 / 3 (43%)

2024 VCAA Exam 2 · Question 11

a.i. Complete the life-cycle (transition) diagram

☐ Birth-rate arrows back to the first group and survival arrows to the next group, all with arrows

Answer / notes: Every edge needed an arrow, or it isn't a directed graph.

State average 0.3 / 1 (30%); 24% scored full marks. Mark distribution: 0: 76% · 1: 24%

Your mark: ☐ **/ 1**

a.ii. Population of each age group after one year

☐ 124, 28, 56, 45

Answer / notes: $L \times R_0$: 0.4(70) + 0.75(80) + 0.4(90) = 124; 0.4(70) = 28; 0.7(80) = 56; 0.5(90) = 45.

State average 0.6 / 1 (60%); 61% scored full marks. Mark distribution: 0: 39% · 1: 61%

Your mark: ☐ **/ 1**

b. Years until the total female population is first half its initial size

☐ 5 years

Answer / notes: Generate $R_n = L^n R_0$ until the total first falls below 140: after 5 years.

State average 0.4 / 1 (40%); 39% scored full marks. Mark distribution: 0: 61% · 1: 39%

Your mark: ☐ **/ 1**

Question 12 (3 marks) · state average 1.2 / 3 (40%)

2023 VCAA Exam 2 · Question 13

a. Minimum distance from G to M

☐ 5.9 km

Answer / notes: 5.9 km. 6.4 km was a common error; some wrote the path without the distance.

State average 0.7 / 1 (70%); 71% scored full marks. Mark distribution: 0: 30% · 1: 71%

Your mark: ☐ **/ 1**

b. A route visiting every landmark and returning to G with minimum distance

☐ G–H–K–I–J–M–O–L–N–G (or the reverse)

Answer / notes: A Hamiltonian cycle. Many missed or misplaced a landmark, or didn't finish at G.

State average 0.4 / 1 (40%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

Your mark: ☐ **/ 1**

c. The two roads travelled twice to cover every road, starting at G and ending elsewhere

☐ L and N, and J and M

Answer / notes: Repeat roads to pair up odd vertices, leaving G and the end vertex odd: roads L–N and J–M. Only 12% got both.

State average 0.1 / 1 (10%); 12% scored full marks. Mark distribution: 0: 88% · 1: 12%

Your mark: ☐ **/ 1**

Question 13 (3 marks) · state average 1.3 / 3 (43%)

2024 VCAA Exam 2 · Question 14

a. Capacity of the marked cut

☐ 46

Answer / notes: Add only edges crossing from the manufacturer side to the supermarket side: 46.

State average 0.8 / 1 (80%); 78% scored full marks. Mark distribution: 0: 22% · 1: 78%

Your mark: ☐ **/ 1**

b. Maximum deliveries per day

☐ 37

Answer / notes: The minimum cut is 37 (by min-cut or by exhausting paths).

State average 0.3 / 1 (30%); 29% scored full marks. Mark distribution: 0: 71% · 1: 29%

Your mark: ☐ **/ 1**

c. Pair of locations whose link should be increased

☐ R and S

Answer / notes: R to S is the bottleneck in the minimum cut. Many other pairs were given.

State average 0.2 / 1 (20%); 22% scored full marks. Mark distribution: 0: 78% · 1: 22%

Your mark: ☐ **/ 1**

Question 14 (2 marks) · state average 0.9 / 2 (45%)

2025 VCAA Exam 2 · Question 16

Values of x, y and z

☐ x = 2, y = 4, z = 2 (one mark for two correct)

Answer / notes: Count passages on the map for each pair. y was often given as 3.

State average 0.9 / 2 (45%); 20% scored full marks. Mark distribution: 0: 32% · 1: 48% · 2: 20%

Your mark: ☐ **/ 2**

Question 15 (4 marks) · state average 1.7 / 4 (43%)

2025 VCAA Exam 2 · Question 18

a. Activities common to both critical paths

☐ A, D, L

Answer / notes: The critical paths are A–D–H–J–L and A–D–I–K–L.

State average 0.5 / 1 (50%); 44% scored full marks. Mark distribution: 0: 56% · 1: 44%

Your mark: ☐ **/ 1**

b. Latest start time for E

☐ 9 days

Answer / notes: 20 − 4 − 4 − 3 = 9.

State average 0.5 / 1 (50%); 46% scored full marks. Mark distribution: 0: 54% · 1: 46%

Your mark: ☐ **/ 1**

c. Activity with the longest float

☐ F

Answer / notes: F (float 6). Extra information had to be correct to keep the mark.

State average 0.5 / 1 (50%); 47% scored full marks. Mark distribution: 0: 53% · 1: 47%

Your mark: ☐ **/ 1**

d. Minimum extra cost to finish three days sooner

☐ $1400

Answer / notes: From 25 to 22 days: reduce A by 2, H by 1 and K by 1 so both critical paths shorten: $1400.

State average 0.2 / 1 (20%); 21% scored full marks. Mark distribution: 0: 79% · 1: 21%

Your mark: ☐ **/ 1**

Exam 1: ☐ **/ 40** **Exam 2:** ☐ **/ 60** **Total:** ☐ **/ 100**

State average on these exact questions: Exam 1 26.6 / 40, Exam 2 30.9 / 60, total 57.5 / 100 (58%). The notes paraphrase the VCAA examiners' reports; read the full reports for sample answers.